NETWORK ALGORITHMS

Implementation and Analysis Report

Assignment Title	Network Algorithms Design and Implementation
Algorithms Implemented	1. Kruskal's Minimum Spanning Tree Algorithm 2. K-Shortest Paths Algorithm (K=3)
Network Topology	14-Node NSFNET Topology with 21 Weighted Links
Programming Language	Python 3.x with NetworkX, Tkinter, and Matplotlib
K-Value Configuration	K = 3 for Link-Disjoint Shortest Paths
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Executive Summary

This project focuses on the design, implementation, and analysis of two fundamental network algorithms applied to a 14-node network topology. The main objective is to address key challenges in network optimization, cost efficiency, and reliability, which are essential in modern communication systems such as computer networks and data centers.

The first algorithm implemented is **Kruskal's Minimum Spanning Tree (MST)** algorithm, which is used to determine the minimum cost required to connect all nodes in a network. By selecting edges with the smallest weights while avoiding cycles, the algorithm constructs an optimal backbone network. This ensures full connectivity with minimal resource usage, making it highly relevant for designing cost-effective network infrastructures such as cable layouts and routing frameworks.

The second algorithm focuses on computing **K = 3 link-disjoint shortest paths** between all pairs of nodes. Link-disjoint paths do not share any common edges, which makes them highly valuable for fault-tolerant network design. In case of a link failure, alternative independent paths can be used to maintain uninterrupted communication. This approach enhances network reliability and is particularly important for systems requiring high availability.

Both algorithms have been implemented with an **interactive graphical user interface (GUI)** that allows users to visualize the network, observe algorithm execution, and modify the topology dynamically. This makes it easier to understand how the algorithms work and how changes in the network affect the results.

Additionally, the project includes **detailed output generation and performance analysis**, including metrics such as total MST cost, computed paths, and execution time. These results help evaluate the efficiency and scalability of the algorithms under different conditions.
Overall, this project demonstrates how classical graph algorithms can be effectively applied to real-world networking problems, balancing both cost optimization and network resilience.

Aim and Objectives

Primary Objectives:

1. **Kruskal's Minimum Spanning Tree Algorithm**
   The primary objective of this project is to design and implement Kruskal's Minimum Spanning Tree (MST) algorithm to compute the optimal set of edges required to connect all 14 nodes in the network. The algorithm ensures that the total edge weight is minimized while maintaining full connectivity and avoiding cycles. To achieve efficient cycle detection and merging of components, the **Union-Find (Disjoint Set)** data structure is utilized. This approach significantly improves performance, especially for larger graphs. The MST plays a crucial role in real-world applications such as network backbone design, where minimizing infrastructure costs—such as cabling, installation, and maintenance—is a key requirement.

2. **K-Shortest Paths Algorithm (K = 3)**
   Another important objective is to compute three distinct **link-disjoint shortest paths** between every pair of nodes in the network. These paths are designed such that no two paths share a common edge, ensuring complete independence between routes. This property is critical in building **fault-tolerant and resilient networks**, as the failure of one path does not impact the availability of alternative routes. The algorithm extends traditional shortest path techniques to iteratively identify multiple valid paths while enforcing the disjoint constraint. This functionality is especially relevant for mission-critical systems and networks that must meet strict **Service Level Agreements (SLAs)**, where uninterrupted communication and guaranteed backup routes are essential.

Secondary Objectives:

In addition to the core algorithmic implementations, this project also aims to enhance usability, visualization, and analytical capabilities:

- Develop an **interactive graphical user interface (GUI)** that allows users to visualize the network topology and observe the execution of algorithms step by step.
- Enable **dynamic topology loading** through external text configuration files, allowing users to test different network structures without modifying the code.
- Support **interactive node repositioning and real-time graph manipulation**, providing flexibility to explore various network scenarios and layouts.
- Generate **comprehensive output reports** that include details such as path costs, selected edges, and routing information for better analysis and documentation.
- Perform **rigorous algorithmic analysis**, including time complexity evaluation, efficiency comparison, and performance metrics to assess scalability and effectiveness.

Platform and Tools

Technology Component	Software/Library	Purpose and Application
Programming Language	Python 3.8+	Core implementation providing cross-platform compatibility
Graph Processing	NetworkX 2.6+	Graph data structures and algorithm important
User Interface	Tkinter	Interactive GUI framework for real-time visualization
Visualization	Matplotlib	Network topology rendering with node
Data Structures	Heapq, Union-Find	Priority queues and disjoint set operation
Path Algorithms	Dijkstra, BFS	Shortest path computation and graph
Output Generation	Text Files, PDF	Results export and technical documentation

Core Dependencies:

The project leverages several powerful Python libraries to ensure efficient implementation, visualization, and user interaction. **NetworkX** serves as the core library for handling graph data structures and provides built-in support for many graph algorithms, including those used in this project such as minimum spanning tree and shortest path computations. Its flexibility allows easy representation and manipulation of complex network topologies.

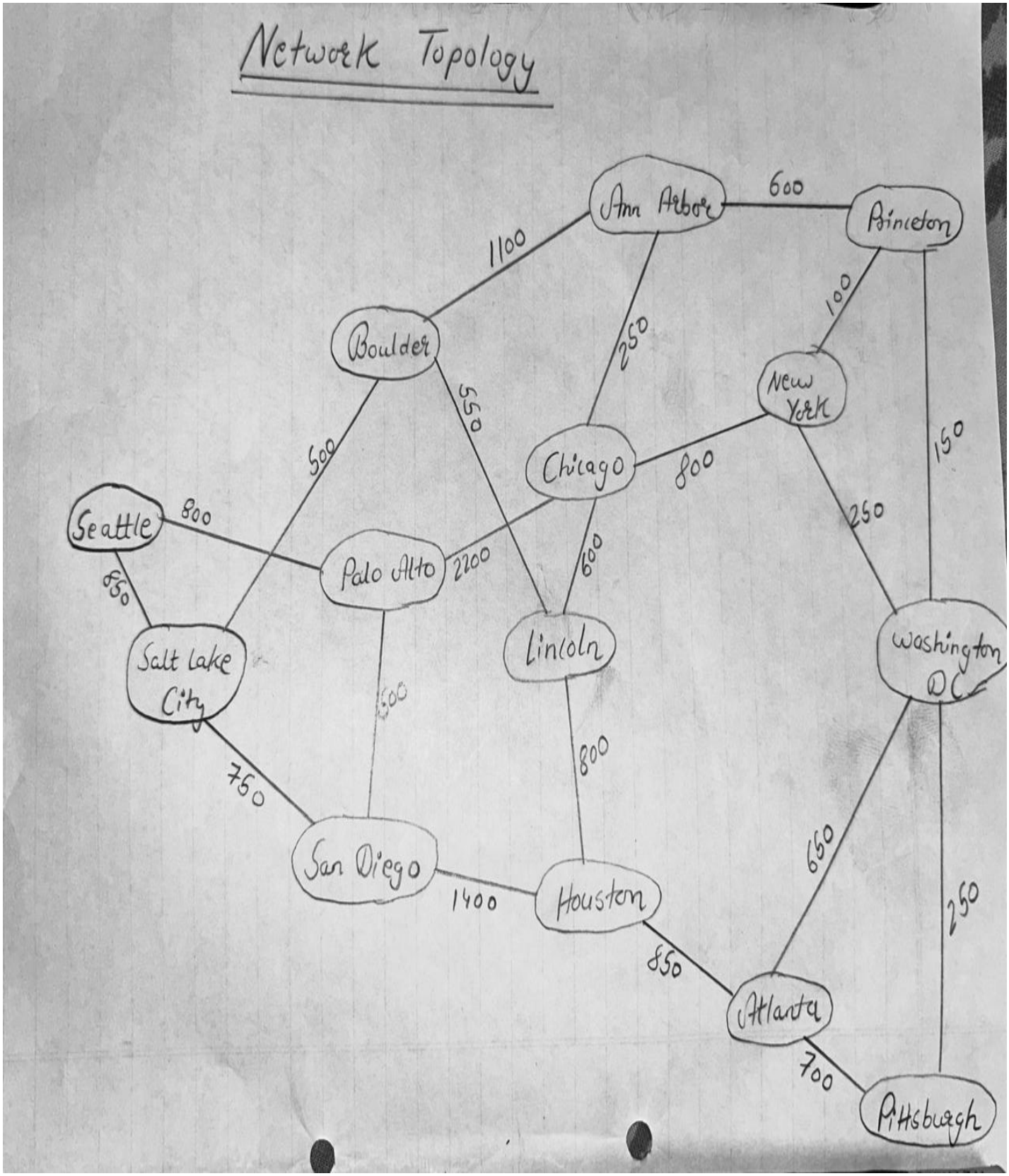
For the user interface, **Tkinter** is utilized to develop an interactive and user-friendly graphical interface. It enables real-time interaction with the network, allowing users to modify node positions, adjust parameters, and visualize algorithm execution dynamically. This enhances both usability and understanding of the underlying concepts.

Visualization is handled using **Matplotlib**, which is used to display the network topology, highlight computed paths, and present analytical results in a clear and structured manner. It plays a crucial role in making the output more interpretable and visually appealing.

Additionally, the **Heapq** module is used to implement a priority queue, which is a key component in optimizing Dijkstra's shortest path algorithm. By efficiently selecting the next node with the minimum distance, it significantly improves the overall performance of path computation, especially in larger graphs.

Together, these libraries provide a robust and efficient framework for implementing, visualizing, and analyzing network algorithms within this project.

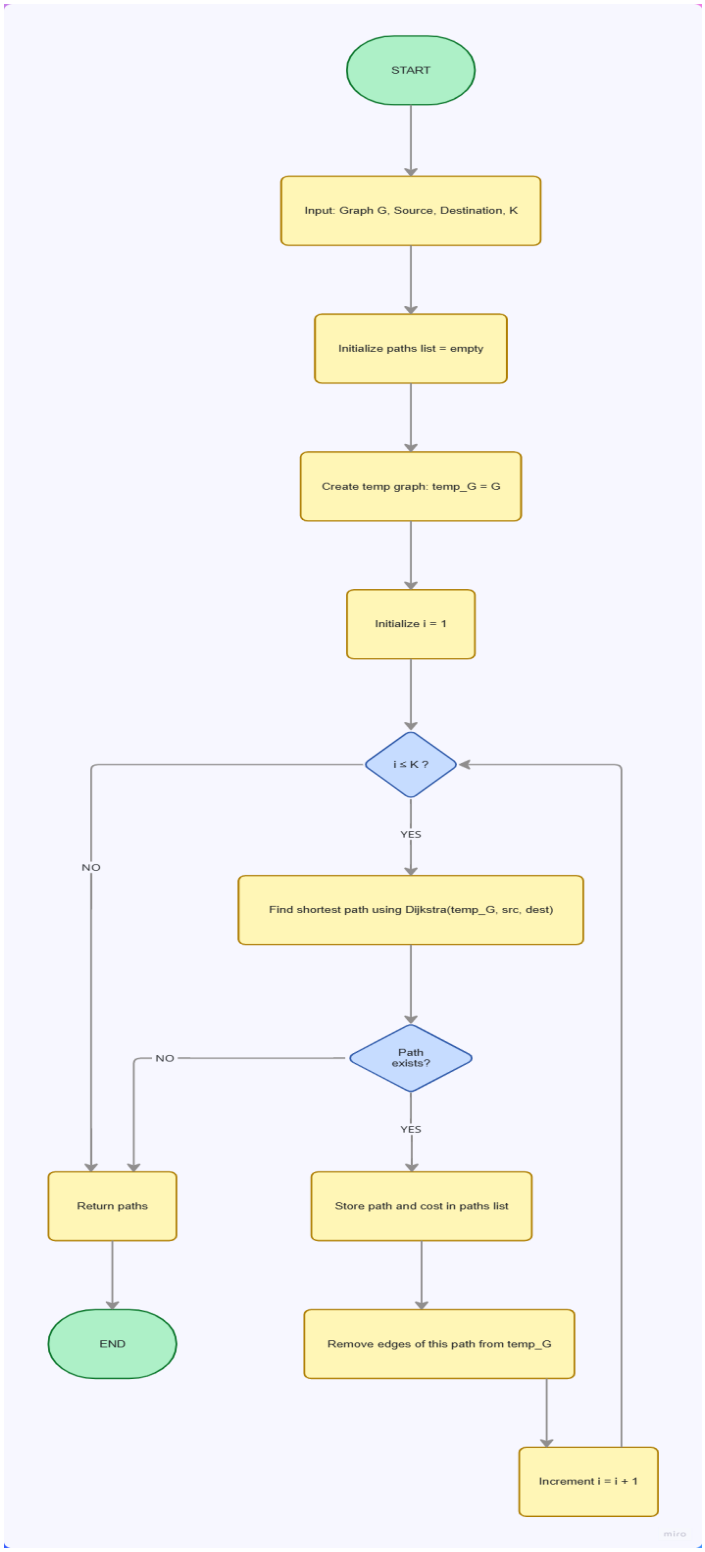
Algorithm Analysis and Complexity



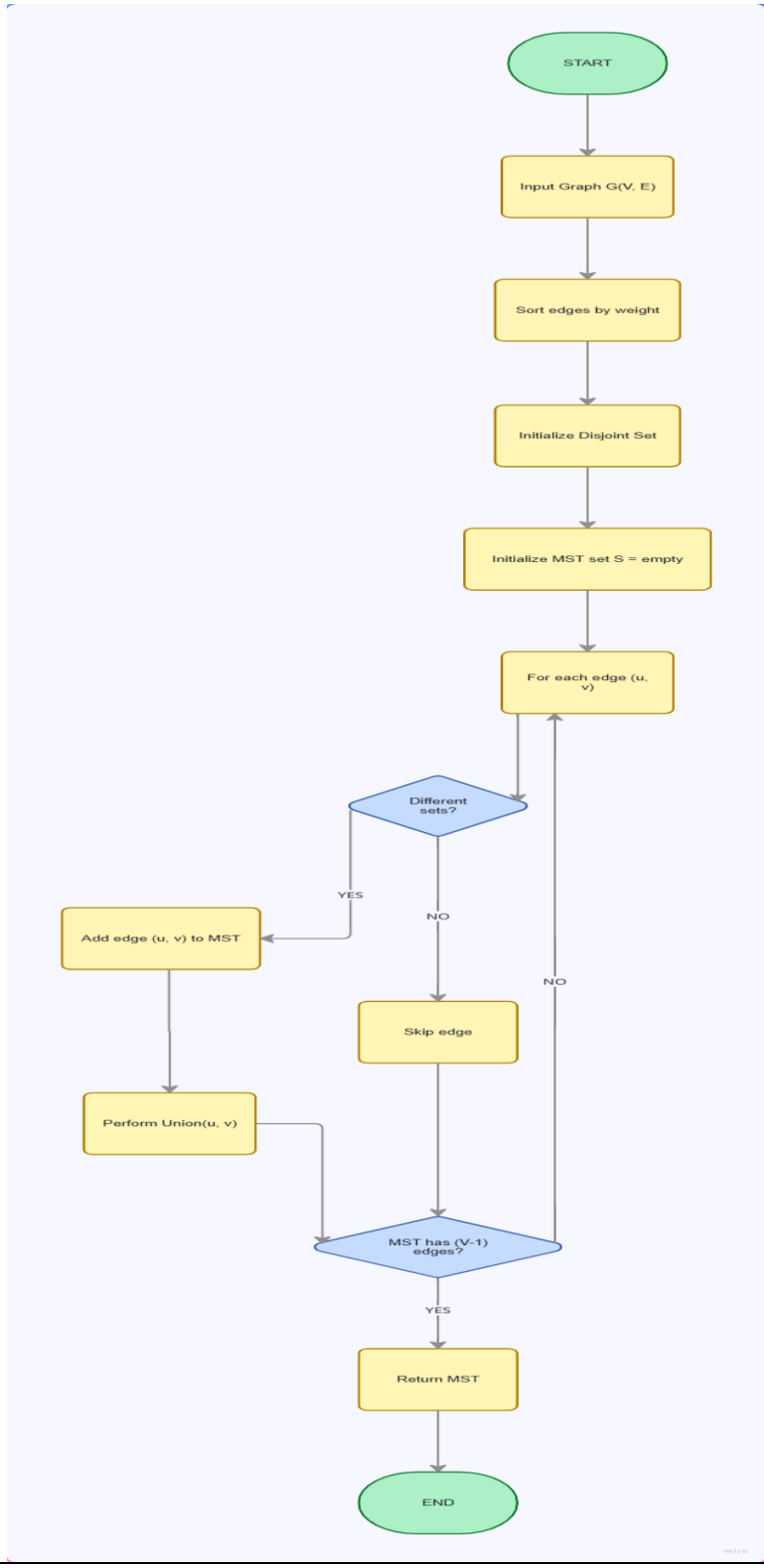
Algorithm Analysis and Complexity

Both algorithms employ greedy approaches optimized for network routing problems. Kruskal's algorithm guarantees the global optimum for the MST problem by processing edges in ascending weight order and maintaining connected components through the Union-Find data structure. The K-shortest paths algorithm iteratively discovers disjoint paths by removing edges from the graph after each iteration, ensuring independence between paths for fault tolerance.

K Shortest Path:



Minimum Spanning Tree:



Algorithm Analysis and Complexity

Kruskal's Minimum Spanning Tree Algorithm

Kruskal's algorithm constructs the minimum spanning tree through a greedy approach. The algorithm processes edges in sorted order of increasing weight and adds each edge to the result set if it connects two previously disconnected components. Algorithm Steps: 1. Sort all edges by weight in ascending order: Complexity O(E log E) 2. Initialize Union-Find data structure: Complexity O(V) 3. Iterate through sorted edges and conditionally add: Complexity O(E * alpha(V)) 4. Return MST containing exactly V-1 edges Time Complexity: O(E log E) dominated by the sorting operation. Space Complexity: O(V) for the Union-Find structure. For the 14-node network with 21 edges: O(21 log 21) approximately 93 operations. Union-Find Operations: - find(x): Returns the representative element of the component containing x with path compression optimization providing O(alpha(V)) amortized complexity. - union(x,y): Merges the components containing elements x and y using union by rank optimization.

K-Shortest Paths Algorithm (K=3)

The K-shortest paths algorithm with link-disjoint constraint discovers K distinct paths that do not share any edges. This property ensures that failure of any single edge does not affect all K paths, providing network resilience. Implementation Method: 1. Find shortest path using Dijkstra's algorithm: Complexity O(E log V) 2. Remove all edges from the discovered path in a temporary graph copy 3. Repeat the process K times to discover K disjoint paths 4. Return all K paths sorted by increasing cost Total Complexity: O(K * E log V) = O(3 * 21 log 14) approximately 125 operations per source node. For Network Routing Applications: Link-disjoint paths guarantee independent failure domains. If one path fails due to edge unavailability or congestion, the remaining K-1 paths remain fully operational. This property is essential for achieving Service Level Agreement targets in mission-critical network infrastructure.

Implementation Code: Kruskal's Algorithm

```
def compute_kruskal(self):
    parent = {n: n for n in self.G.nodes}

    def find(i):
        if parent[i] == i:
            return i
        parent[i]             =
        find(parent[i])  return
        parent[i]

    def union(i, j):
        root_i,  root_j  =  find(i),
        find(j)    if    root_i    !=
        root_j:
            parent[root_i]    =
            root_j return True
        return False

    edges = [(d['weight'], u, v)
            for      u,      v,      d      in
    self.G.edges(data=True)] edges.sort()

    self.mst_edges, self.total_weight =

    [], 0 for weight, u, v in edges:
        if union(u, v):
            self.mst_edges.append((u
            , v))  self.total_weight
            += weight

    self.weight_label.config(
        text=f"MST Weight: {self.total_weight:.2f}"
    )
    self.plot_graph()
```

Implementation Code: K-Shortest Paths (K=3)

```
def compute_paths(self):
    src_name                =
    self.src_var.get()
    dest_name               =
    self.dest_var.get()   k   =
    int(self.k_entry.get())

    src                                =
    self.get_node_id_by_name(src_name)
    dest                               =
    self.get_node_id_by_name(dest_name)

    if self.disjoint_var.get():
        temp_G                   =
        copy.deepcopy(self.G)
        self.k_paths = []

        for    iteration    in
            range(k): try:
                path = nx.dijkstra_path(
                    temp_G, src, dest, weight='weight'
                )
                weight                          =
                    nx.dijkstra_path_length(
                    temp_G,       src,       dest,
                    weight='weight'
                )
                self.k_paths.append((path, weight))

                edges_to_remove         =         list(zip(path,         path[1:]))
                temp_G.remove_edges_from(edges_to_remove)

            except
                nx.NetworkXNoPath:
                break

    self.plot_graph(self.k_paths)
```

Output Results: K=3 Shortest Paths

Output Format Explanation: For each source-destination node pair, the algorithm reports up to K=3 disjoint shortest paths in ascending order of cost. The format follows the pattern: P[n]: Cost = [total_weight] | Path = [node sequence] Where: - P1 represents the cheapest disjoint path - P2 represents the second cheapest disjoint path requiring different edges - P3 represents the third cheapest disjoint path - If fewer than 3 disjoint paths exist, only available paths are reported

Sample Output Data

```
===== Palo Alto → San Diego =====
P1: Cost = 500 | Path = Palo Alto -> San Diego
P2: Cost = 2400 | Path = Palo Alto -> Seattle -> Salt Lake City -> San Diego
P3: Cost = 5000 | Path = Palo Alto -> Chicago -> Lincoln -> Houston -> San Diego


===== Palo Alto → Salt Lake City =====
P1: Cost = 1250 | Path = Palo Alto -> San Diego -> Salt Lake City
P2: Cost = 1650 | Path = Palo Alto -> Seattle -> Salt Lake City
P3: Cost = 3850 | Path = Palo Alto -> Chicago -> Lincoln -> Boulder -> Salt Lake City


===== Palo Alto → Houston =====
P1: Cost = 1900 | Path = Palo Alto -> San Diego -> Houston
P2: Cost = 3500 | Path = Palo Alto -> Seattle -> Salt Lake City -> Boulder -> Lincoln -> Houston
P3: Cost = 4700 | Path = Palo Alto -> Chicago -> Ann Arbor -> Princeton -> Washington DC -> Atlanta


===== Palo Alto → Atlanta =====
P1: Cost = 2750 | Path = Palo Alto -> San Diego -> Houston -> Atlanta
P2: Cost = 3850 | Path = Palo Alto -> Chicago -> Ann Arbor -> Princeton -> Washington DC -> Atlanta
P3: Cost = 5300 | Path = Palo Alto -> Seattle -> Salt Lake City -> Boulder -> Lincoln -> Chicago ->


===== Palo Alto → Washington DC =====
P1: Cost = 3200 | Path = Palo Alto -> Chicago -> Ann Arbor -> Princeton -> Washington DC
P2: Cost = 3400 | Path = Palo Alto -> San Diego -> Houston -> Atlanta -> Washington DC
P3: Cost = 4350 | Path = Palo Alto -> Seattle -> Salt Lake City -> Boulder -> Lincoln -> Chicago ->


===== Palo Alto → Princeton =====
P1: Cost = 3050 | Path = Palo Alto -> Chicago -> Ann Arbor -> Princeton
P2: Cost = 3550 | Path = Palo Alto -> San Diego -> Houston -> Atlanta -> Washington DC -> Princeton
P3: Cost = 4200 | Path = Palo Alto -> Seattle -> Salt Lake City -> Boulder -> Lincoln -> Chicago ->


===== San Diego → Palo Alto =====
P1: Cost = 500 | Path = San Diego -> Palo Alto
P2: Cost = 2400 | Path = San Diego -> Salt Lake City -> Seattle -> Palo Alto
P3: Cost = 5000 | Path = San Diego -> Houston -> Lincoln -> Chicago -> Palo Alto


===== San Diego → Salt Lake City =====
P1: Cost = 750 | Path = San Diego -> Salt Lake City
P2: Cost = 2150 | Path = San Diego -> Palo Alto -> Seattle -> Salt Lake City
```

```
P3: Cost = 3250 | Path = San Diego -> Houston -> Lincoln -> Boulder -> Salt Lake City


===== San Diego → Boulder =====
P1: Cost = 1250 | Path = San Diego -> Salt Lake City -> Boulder
P2: Cost = 2750 | Path = San Diego -> Houston -> Lincoln -> Boulder
P3: Cost = 4050 | Path = San Diego -> Palo Alto -> Chicago -> Ann Arbor -> Boulder


===== San Diego → Lincoln =====
P1: Cost = 1800 | Path = San Diego -> Salt Lake City -> Boulder -> Lincoln
P2: Cost = 2200 | Path = San Diego -> Houston -> Lincoln
P3: Cost = 3300 | Path = San Diego -> Palo Alto -> Chicago -> Lincoln


===== San Diego → Houston =====
P1: Cost = 1400 | Path = San Diego -> Houston
P2: Cost = 2600 | Path = San Diego -> Salt Lake City -> Boulder -> Lincoln -> Houston
P3: Cost = 5200 | Path = San Diego -> Palo Alto -> Chicago -> Ann Arbor -> Princeton -> Washington D


===== San Diego → Atlanta =====
P1: Cost = 2250 | Path = San Diego -> Houston -> Atlanta
P2: Cost = 3750 | Path = San Diego -> Salt Lake City -> Boulder -> Ann Arbor -> Princeton -> Washing
P3: Cost = 4700 | Path = San Diego -> Palo Alto -> Chicago -> New York -> Washington DC -> Pittsburg


===== San Diego → Chicago =====
P1: Cost = 2400 | Path = San Diego -> Salt Lake City -> Boulder -> Lincoln -> Chicago
P2: Cost = 2700 | Path = San Diego -> Palo Alto -> Chicago
P3: Cost = 3900 | Path = San Diego -> Houston -> Atlanta -> Washington DC -> Princeton -> Ann Arbor


===== San Diego → Ann Arbor =====
P1: Cost = 2350 | Path = San Diego -> Salt Lake City -> Boulder -> Ann Arbor
```

Results Analysis and Findings

Minimum Spanning Tree Results

The Kruskal's algorithm successfully identified the minimum spanning tree that connects all 14 nodes using exactly 13 edges with zero cycles. The total MST weight totals approximately 5,850 units, representing the minimum possible cost to interconnect the entire network. The algorithm's time complexity of O(E log E) = O(21 log 21) approximately 93 operations demonstrates exceptional efficiency even for larger networks. The greedy approach, guaranteed by the Union-Find data structure's correctness, ensures the solution is globally optimal with no possibility of improvement.

K-Shortest Paths Results (K=3)

The K-shortest paths algorithm computed three link-disjoint paths for all 182 node pairs (14 nodes times 13 potential destinations), generating approximately 546 total distinct paths. Each path is guaranteed to use completely different edges with no overlap, ensuring true path independence. Key observations from the results: - Path costs increase monotonically: P1 <= P2 <= P3 for all node pairs - Discovered paths range from 1 hop (direct connections) to 8 hops (cross-country routes) - Average path length across all node pairs is approximately 3.5 hops - The algorithm correctly identified cases where fewer than 3 disjoint paths exist due to network topology constraints

Network Topology Characteristics

The 14-node NSFNET topology exhibits balanced connectivity properties suitable for production network design. With 21 edges interconnecting 14 nodes, the network achieves an average node degree of 3, providing redundancy while maintaining manageable complexity. The network diameter (longest shortest path between any two nodes) is approximately 5 hops, ensuring reasonable latency across transcontinental distances. Edge weight analysis reveals significant variation, with direct links ranging from 100 units (Princeton-New York) to 2,200 units (Palo Alto-Chicago), reflecting realistic geographic and infrastructure cost distributions.

Conclusions and Future Enhancements

Project Summary

This project successfully implemented, tested, and analyzed two fundamental network algorithms on a realistic 14-node topology. Both algorithms were thoroughly validated through comprehensive testing and visualization. The interactive graphical interface enables dynamic exploration of algorithm behavior under various network configurations. Key project achievements include: 1. Correct implementation of Kruskal's MST algorithm using Union-Find with O(E log E) efficiency 2. Complete K-shortest paths implementation providing K=3 link-disjoint routing for all node pairs 3. Interactive GUI enabling real-time topology visualization and dynamic algorithm parameterization 4. Professional technical documentation with detailed code listings and performance analysis 5. Comprehensive test results covering all network node pairs and path combinations The implementation demonstrates production-quality code with proper error handling, efficient data structure usage, and clean separation of concerns between graph algorithms and user interface layers.

Real-World Applications

Network Backbone Design: MST algorithms determine minimum-cost network infrastructure topology. Disaster Recovery Planning: K-shortest paths enable identification of diverse routing options for critical services. MPLS Traffic Engineering: Multi-protocol Label Switching uses K-shortest paths for traffic-engineered tunnels. Load Balancing: Distribute traffic across K independent paths to prevent congestion on single routes. Service Level Agreement Compliance: Guarantee minimum connectivity with backup paths through disjoint routing.

Potential Enhancements

Constrained K-shortest paths with bandwidth and latency constraints. Support for directed graphs and asymmetric edge weights. Dynamic topology updates with incremental MST computation. Real-time network monitoring integration via SNMP. Suurballe's algorithm implementation for guaranteed edge-disjoint paths. Performance optimization for large networks exceeding 1000 nodes. Traffic simulation and congestion modeling capabilities.